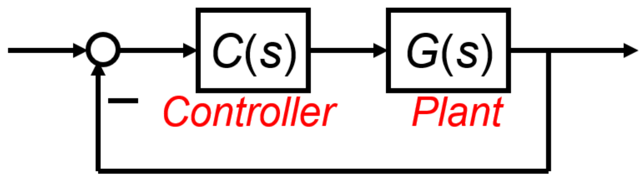
Assignment 15 (ELEC 341 L15_RLPID)

Problem 1:

Determine the Ziegler-Nichols tuning parameters for a PID controller, with the given plant transfer function:

$$G(s) = \left(\frac{5}{2s+1}\right)\left(\frac{0.4}{5s+1}\right)\left(\frac{2}{s+1}\right) = \frac{4}{(2s+1)(5s+1)(s+1)} = \frac{4}{10s^3+17s^2+8s+1}$$

$(2s+1)(5s+1)$
$= 10s^2+2s+5s+1$
$\rightarrow (10s^2+7s+1)(s+1)$
$= 10s^3+7s^2+s+10s^2+7s+1$
$= 10s^3+17s^2+8s+1$



Assume that the time constants have units of minutes and the controller transfer function $C(s)$ is as follows:

$$C(s) = K_p\left(1 + \frac{1}{T_I s} + T_D s\right)$$

$K_p = 0.6K_c$
$T_i = 0.5K_c$
$T_d = 0{,}125K_c$

Use Matlab to graph the step response.

let $k_i, k_d = 0$. and let $K_p = K_c$

$C(s) = K_c$, $L(s) = C(s)G(s)$

$1 + L(s) = 0$.

$$1 + \frac{4K_c}{10s^3+17s^2+8s+1} = 0.$$

$10s^3+17s^2+8s+1+4K_c = 0$.

s^3	10	8
s^2	17	$1+4K_c$
s^1	$\frac{126-40K_c}{17}$	0
s^0	$1+4K_c$	

$\frac{136-10-40K_c}{17} = \frac{126-40K_c}{17} > 0$

$126 < 40K_c$

$K_c < 3{,}15$

at $K_c = 3{,}15$

$P(s) = 17s^2 + 13{,}6 = 0$,

$17s^2 = -13{,}6$

$s = \pm j\sqrt{0{,}8}$

$s = \pm j\frac{2}{\sqrt{5}}$

$\omega = \frac{2\pi}{T_c} \Rightarrow \boxed{T_c = \sqrt{5}\,\pi \text{ min}}$

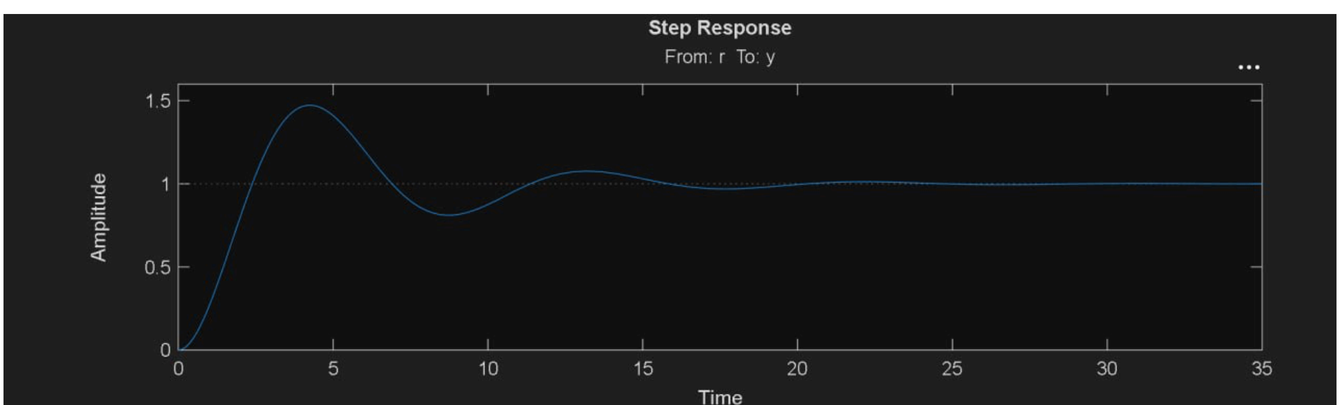
$K_p = 0{,}6K_c$
$K_p = (0{,}6)(3{,}15)$
$\boxed{K_p = 1.89}$

$T_i = 0{,}5T_c$
$T_i = \frac{\sqrt{5}\pi}{2}$
$\boxed{T_i = 3.5122 \text{ min}}$

$T_d = 0{,}125T_c$
$\boxed{T_d = 0.8781 \text{ min}}$

$C(s) = K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$

$\boxed{C(s) = 1{,}89\left(1 + \frac{1}{3.5122s} + 0.8781s\right)}$



minutes.